

YOUSSEF BASTAWISY

AI Engineer · Egypt

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Summary

AI Engineer specializing in production-ready **agentic AI systems**—multi-agent orchestration, Retrieval-Augmented Generation (RAG), structured outputs, and function calling / tool-use. Builds autonomous, stateful, tool-aware agents end-to-end with **Pydantic AI, LlamaIndex, LangGraph, FastAPI, and Streamlit**, and automates them into real-world workflows with **n8n**. Ships reliable, observable LLM products from prototype to live, production-grade services, with strong foundations in Python, machine learning, and token-efficient agent design.

Experience

Kayfa Academy

Jun 2026 – Present

Agentic AI Instructor (Part-time)

Cairo, Egypt

- Deliver hands-on training in Agentic AI, covering RAG, multi-agent systems, AI workflows, and production deployment.
- Mentor learners in building and deploying AI-powered applications and autonomous agents.

Restart Technology (Laam)

Apr 2026 – Present

AI Engineer

Egypt

- Build and ship production LLM systems end-to-end—RAG pipelines, function calling and tool-use, model serving, vector databases, deployment, observability, and reliability.
- Take AI agents from prototype to live, production-grade services owning the full lifecycle.

Cycls

Aug 2025 – Present

Applied AI Engineer

Egypt

- Build and ship LLM-powered conversational and agentic applications on the company's platform, taking them from prototype to production.
- Implement function calling and tool-use to connect agents with external APIs and data sources, and deploy to scalable serverless infrastructure (Modal) with a focus on reliability, fast iteration, and end-user experience.

Arabian Academy

Aug 2025 – Present

Data Science Instructor

Egypt

- Deliver hands-on training in Python, data analysis, and machine learning, and design real-world, project-based curricula that build industry-ready, data-driven problem-solving skills.

ZA Tech

Jul 2024 – Sep 2024

Machine Learning Intern

Egypt

- Designed and evaluated predictive models for insurance and fintech use cases; preprocessed real-world datasets and optimized performance via cross-validation and hyperparameter tuning, deploying models with Python and cloud tools.

WorldQuant University

Jan 2023 – 2024

Data Science Intern

Remote

- Built end-to-end ML pipelines—data cleaning, feature engineering, model development, and evaluation—using Pandas, scikit-learn, and TensorFlow across finance and health datasets.

PES – Programmers Elite School

Sep 2022 – Mar 2023

Data Science Instructor

Egypt

- Taught Python, data analysis, and machine learning through live coding sessions and hands-on projects; designed lesson plans and mentored students on real-world datasets.

Projects

Laam — Arabic RAG Assistant | *RAG, vLLM, TEI Embeddings, Qdrant, FastAPI, Caddy, Tesseract OCR* **May 2026**

- Built and deployed an end-to-end Arabic RAG assistant grounded in the platform's knowledge base, engineering the retrieval pipeline (semantic chunking, embeddings, cross-encoder reranking, deduplication) to reach hit@8 of 0.96 on the evaluation set.
- Self-hosted the full serving stack (vLLM-served LLM, TEI embeddings, Qdrant vector DB) on an NVIDIA Tesla T4 on Google Cloud, streaming Arabic responses (SSE) over HTTPS behind Caddy; built a Tesseract OCR ingestion pipeline converting 47 Ministry-of-Education PDFs into 2,800 indexed chunks.
- Added JWT-secured function calling for live reads from the platform's Laravel API, plus production observability, automated daily vector-store backups, and a HubSpot human-handoff bridge.

Super Agent | *Cycls SDK, Agentic Orchestration, Code Execution, Web Search, RAG*

Mar 2026

- Built a general-purpose agentic loop with isolated code execution, file creation/editing, and web search, spanning research, business, and reasoning tasks.
- Integrated RAG and deep-search components into the orchestrator so the agent can plan, retrieve, and act autonomously across multi-step objectives.

Bid Evaluation Expert | *Cycls SDK, Claude Sonnet 4.6, Function Calling, Structured Output, openpyxl* **Jan 2026**

- Built a structured-output, function-calling agent that evaluates technical and financial bids using a weighted scoring formula validated to sum to 100%, producing a separate documented evaluation per approved standards.
- Designed a typed `generate_excel_template` tool that emits a ready-to-fill, multi-sheet Excel workbook covering the five official local-content mechanisms, with a composable evaluation-skills layer injected into the system prompt.

TasiBot — Saudi Market Analyst | *Cycls SDK, Claude Skills, Live Data, Chart.js / QuickChart* **Dec 2025**

- Built an agentic stock analyst covering 400+ TASI-listed stocks with live quotes, technical signals, scoring, gainers/losers, dividends, social sentiment, watchlists, and breaking disclosures.
- Implemented a `render_chart` tool producing interactive bar/line/pie/doughnut charts (via QuickChart), a Skills-based market-data layer, and an AskUserQuestion clarifying tool for reliable, friendly interactions.

Haseef — Saudi Legal Assistant | *Cycls SDK, Text-to-SQL RAG, Postgres FTS, Tool Use* **Oct 2025**

- Built a specialized legal assistant retrieving over a structured Postgres legal database (laws + court cases) through a guarded read-only SQL tool (SELECT/WITH only, row caps and statement timeouts, public views only).
- Implemented always-ranked full-text search with payload capping and file-spill so large law/case bodies never bloat context, plus Hijri/Gregorian date handling and per-workspace template sync.

Local Content Expert | *Cycls SDK, Source-Index RAG, Structured UI Tools, Sandbox Tools* **Aug 2025**

- Built an Arabic-first agent answering local-content regulatory questions via index-first RAG over a curated source index, consulting the index before exploratory search to keep retrieval grounded and accurate.
- Engineered per-turn conversation-history sanitization to stabilize retrieval behavior and shrink context, with rich table / step / callout rendering through JSON-Schema structured UI tools.

Technical Skills

Languages: Python, R, SQL

Agentic & LLM Frameworks: Pydantic AI, LangGraph, LangChain, LlamaIndex, Claude Agent SDK, Cycls SDK, Hugging Face

GenAI / LLMs: RAG, function calling & tool-use, multi-agent orchestration, structured outputs, prompt engineering, embeddings & reranking, semantic search

Retrieval & Serving: Qdrant, ChromaDB, vLLM, TEI, Tesseract OCR, SSE streaming

Backend & MLOps: FastAPI, Streamlit, n8n, Docker, Modal, Google Cloud (GCP), Caddy, REST APIs, Git, Linux

ML & Deep Learning: scikit-learn, TensorFlow, Keras, Transformers, NLTK, SpaCy, CNN/RNN/LSTM

Databases: PostgreSQL/Supabase, SQL Server, MongoDB, Firebase, DynamoDB

Visualization: Matplotlib, Seaborn, Plotly, Power BI, Tableau, Looker Studio

Education

Pharos University in Alexandria

Alexandria, Egypt

B.Sc. in Computer Science & Artificial Intelligence — Major: Data Science GPA: 3.8/4.0

Certificates

IBM Data Science Specialization: End-to-end data science workflow across Python, SQL, data analysis, visualization, and machine learning.

Machine Learning Using SAS Viya 3.5: Supervised and unsupervised modeling and model assessment in SAS Viya.

Intermediate SQL Queries — DataCamp: Advanced querying, aggregation, and data manipulation in SQL.

Data Science & Advanced Analytics Virtual Experience — Forage: Applied analytics simulation covering real-world data science tasks.